

~\Desktop\LAB\sparse.c

```
1 //SPARSE MATRIX AND ITS TRANSPOSE REPRESENTATION
2 #include<stdio.h>
3 void main()
4 {
5     int a[5][5],i,j,k=0,r,c,p=1;
6     printf("enter the order of the matrix \n");
7     scanf("%d%d",&r,&c);
8     printf("the elements are \n");
9     for(i=0;i<r;i++)
10         for(j=0;j<c;j++)
11             scanf("%d",&a[i][j]);
12     for(i=0;i<r;i++)
13         for(j=0;j<c;j++)
14             {
15                 if(a[i][j]!=0)
16                     k++;
17             }
18     int sparse[k+1][3],trans[k+1][3];
19     sparse[0][0]=r;
20     sparse[0][1]=c;
21     sparse[0][2]=k;
22     for(i=0;i<r;i++)
23         for(j=0;j<c;j++)
24             if(a[i][j]!=0)
25             {
26                 sparse[p][0]=i;
27                 sparse[p][1]=j;
28                 sparse[p][2]=a[i][j];
29                 p++;
30             }
31     trans[0][0]=c;
32     trans[0][1]=r;
33     trans[0][2]=k;
34     p=1;
35     for(i=0;i<c;i++)
36         for(j=1;j<k+1;j++)
37             if(sparse[j][1]==i)
38             {
39                 trans[p][0]=sparse[j][1];
40                 trans[p][1]=sparse[j][0];
41                 trans[p][2]=sparse[j][2];
42                 p++;
43             }
```

```
44     printf("sparse representation\n");
45     for(i=0;i<k+1;i++)
46     {
47         for(j=0;j<3;j++)
48         {
49             printf("%d ",sparse[i][j]);
50         }
51         printf("\n");
52     }
53     printf("sparse transpose representation\n");
54     for(i=0;i<k+1;i++)
55     {
56         for(j=0;j<3;j++)
57         {
58             printf("%d ",trans[i][j]);
59         }
60         printf("\n");
61     }
62 }
63
64
65
66
67 /* OUTPUT :
68 enter the order of the matrix
69 3
70 2
71 the elements are
72 1
73 0
74 2
75 0
76 0
77 3
78 sparse representation
79 3 2 3
80 0 0 1
81 1 0 2
82 2 1 3
83 sparse transpose representation
84 2 3 3
85 0 0 1
86 0 1 2
87 1 2 3
88 */
```